



Pimpri Chinchwad Education Trust's  
**Pimpri Chinchwad College of Engineering (PCCoE)**  
(An Autonomous Institute)  
Affiliated to Savitribai Phule Pune University (SPPU)  
ISO 21001:2018 Certified by TUV SUD

**BCE26PE03-UI and UX Design : Question Bank**

**Class : Third Year (All Division)**

**Academic Year: 2025 - 2026**

**Semester: II**

**1. Name of Module: Application Development**

**2. Name of Module Coordinator/Course Reviewer: Dr. Shrinivas Ambala**

**3. Name of Course: PEC - II UI and UX Design (BCER26PE03) Class: TY Div A,B,C,D**

**4. Name of Course Owner: Ms Rohini Tambe (CO)**

**5. Name of Course Faculties: Ms. Deepali Jawale, Dr. Anagha Chaudhari**

| CO No | Course Outcome                                                                                           |
|-------|----------------------------------------------------------------------------------------------------------|
| CO1   | Explore the design thinking process and the distinction between UI and UX in system design.              |
| CO2   | Create user personas and journey maps based on findings from user research.                              |
| CO3   | Apply visual design principles and design systems to develop effective wireframes.                       |
| CO4   | Develop interactive UI prototypes and responsive front-end interfaces using design tools and frameworks. |

**Instructions to Students:**

1. Refer class notes, PPTs, and standard UI/UX reference book materials for preparation shared on google classroom.
2. Focus on key topics: User Research, Personas, Empathy Maps, Task Flows, and User Journey Mapping etc.
3. **Questions are for practice**; similar or new scenario-based questions may appear in exams.
4. Write answers in points with proper headings and relate them to the given scenario.
5. Draw diagrams wherever required (Persona, Empathy Map, Journey, Task Flow etc.).
6. Provide clear and step-by-step explanations in all answers.

**UNIT I:** Introduction: Overview of system design with a focus on UI/UX. Design Thinking Process: Empathize, Define, Ideate, Prototype, Test, Human-centered design principles, Difference between UI and UX, Role of UX in system-level architecture.

| Sr. No | Questions                                                                                                                                                                                                                                        | Difficulty Level | Blooms Level | Course Outcome |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------|----------------|
| 1      | Describe the concepts of User Interface (UI) and User Experience (UX), and distinguish between them by highlighting two key differences with suitable examples from a mobile application.                                                        | Low              | Remember     | CO1            |
| 2      | Identify the five stages of the Design Thinking process and briefly state the goal associated with each stage.                                                                                                                                   | Low              | Remember     | CO1            |
| 3      | Describe the concept of Human-Centered Design (HCD) and discuss any three core principles it follows in digital product design.                                                                                                                  | Low              | Remember     | CO1            |
| c      | Identify any four commonly used UI/UX design tools and describe the primary purpose of each.                                                                                                                                                     | Low              | Remember     | CO1            |
| 5      | Describe the activities performed in the 'Ideate' stage. List two brainstorming techniques used in this stage.                                                                                                                                   | Low              | Understand   | CO1            |
| 6      | Describe the goal of the 'Define' stage in Design Thinking. What deliverable does it typically produce?                                                                                                                                          | Low              | Understand   | CO1            |
| 7      | List the advantages of implementing UX design in digital products.                                                                                                                                                                               | Low              | Understand   | CO1            |
| 8      | Compare User Interface (UI) and User Experience (UX) based on their characteristics. Also explain how UX plays an important role in modern UI design.(Consider aspects like visual design, usability, user satisfaction, and overall experience) | Low              | Understand   | CO1            |
| 9      | List the stages involved in the design thinking process used in UI/UX design                                                                                                                                                                     | Low              | Understand   | CO1            |
| 10     | Examine the concept of user personas and analyze their role in understanding user needs and behaviors in the design process.                                                                                                                     | Low              | Understand   | CO1            |
| 11     | Classify prototypes based on fidelity and functionality characteristics.                                                                                                                                                                         | Medium           | Analyze      | CO1            |

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| 12 | Analyze a healthcare portal in which the interface appears visually appealing, yet elderly patients experience difficulty locating their medical records. Distinguish the UI issue from the UX issue, and propose one appropriate fix for each. | Medium | Analyze | CO1 |
| 13 | Apply the five-stage Design Thinking process to improve the checkout flow of an e-commerce site with a 65% cart abandonment rate, and describe the actions taken at each stage.                                                                 | Medium | Apply   | CO1 |
| 14 | Apply Human-Centered Design principles to a government portal where elderly users struggle with confusing navigation, identify two key problems, and propose suitable improvements for each.                                                    | Medium | Apply   | CO1 |
| 15 | Analyze a task flow diagram for an e-commerce checkout process and identify inefficiencies affecting user completion                                                                                                                            | Medium | Analyze | CO1 |
| 16 | Differentiate between user thoughts, feelings, and actions within an empathy map.                                                                                                                                                               | Medium | Apply   | CO1 |
| 17 | Identify gaps in a user journey map designed for an online food delivery system.                                                                                                                                                                | Medium | Apply   | CO1 |
| 18 | Analyze the negative impact on user engagement when a social media platform is designed solely around business KPIs without incorporating Human-Centered Design principles.                                                                     | Medium | Analyze | CO1 |
| 19 | A public transport app has poor route search and confusing timetable display. Apply the Design Thinking process to propose an improved user experience.                                                                                         | Medium | Apply   | CO1 |
| 20 | Differentiate the impact of UI improvements and UX improvements on employee productivity in the redesign of an ERP interface.                                                                                                                   | Medium | Analyze | CO1 |
| 21 | Apply UX feedback and error recovery principles to enhance a mobile banking app interaction where transaction errors occur without user-facing messages or recovery options.                                                                    | Medium | Apply   | CO1 |
| 22 | Analyze the relationship between touchpoints and user emotions in a user journey map.system, and examine their role in generating user-centered design solutions.                                                                               | Medium | Analyze | CO1 |

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| 23 | Apply task flow diagramming techniques to represent the user journey of booking a cab through a mobile application.                                                                                                                                   | Medium | Apply    | CO1 |
| 24 | A startup launches a product without proper prototyping and user testing, leading to failure. Analyze the importance of prototyping and testing stages in the Design Thinking process and justify how they could have prevented the failure.          | Medium | Analyze  | CO1 |
| 25 | A team struggles to generate innovative solutions during product design. Apply ideation techniques to propose creative solutions and justify how they improve the design outcome.                                                                     | Medium | Apply    | CO1 |
| 26 | Evaluate the effectiveness of user personas in guiding UX decisions in a healthcare application.                                                                                                                                                      | Medium | Evaluate | CO1 |
| 27 | A hospital management system is technically efficient but doctors find it stressful and time-consuming to use. Evaluate the role of UX in system-level architecture and recommend changes to improve overall system efficiency and user satisfaction. | Medium | Evaluate | CO1 |
| 28 | Evaluate the effectiveness of a user journey map in improving service quality in a banking application.                                                                                                                                               | Medium | Evaluate | CO1 |
| 29 | Evaluate the effectiveness of task flow diagrams in improving collaboration between UX designers and developers.                                                                                                                                      | Medium | Evaluate | CO1 |
| 30 | Analyze the relationship between touchpoints and user emotions in a user journey map                                                                                                                                                                  | Medium | Analyze  | CO1 |
| 31 | Evaluate the application of Human-Centered Design principles in guiding both system architecture and interface design for a smart city traffic system serving commuters, police, and city administrators..                                            | High   | Evaluate | CO1 |
| 32 | Design a comprehensive UX strategy using all five stages of the Design Thinking framework to address a 70% onboarding dropout rate in an e-learning platform.                                                                                         | High   | Create   | CO1 |
| 33 | Design a complete UX flow for an investment application targeting first-time investors with low                                                                                                                                                       | High   | Create   | CO1 |

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|    | financial literacy using the Design Thinking framework, and justify key UI and UX decisions made throughout the process.                                                                                                      |      |          |     |
| 34 | Critically evaluate the effectiveness of the Design Thinking process as a problem-solving methodology in UI/UX design projects, and support the evaluation with two real-world examples demonstrating its success or failure. | High | Evaluate | CO1 |
| 35 | Evaluate the role of UX architecture in an airport management portal serving airline staff, ground crew, and passengers, and justify the necessity of integrating UX at the system design level..                             | High | Evaluate | CO1 |
| 36 | Evaluate the design of a healthcare chatbot using Human-Centered Design criteria based on mixed usability feedback, and formulate a revised design strategy to improve the conversational experience.                         | High | Evaluate | CO1 |
| 37 | Evaluate the user experience of a real-world application by identifying usability strengths and limitations, and propose well-justified improvements to enhance overall user satisfaction.                                    | High | Evaluate | CO1 |
| 38 | Design a user-centered ATM interface tailored for elderly and differently-abled users, incorporating accessibility features, usability principles, and inclusive design considerations with appropriate justification.        | High | Evaluate | CO1 |
| 39 | Analyze the situation to distinguish between UI and UX issues and determine why visual improvements alone did not enhance user engagement.                                                                                    | High | Analyze  | CO1 |
| 40 | Evaluate the distinct contributions of UI and UX improvements in enhancing employee productivity during an ERP system redesign, supported with relevant examples                                                              | High | Analyze  | CO1 |

**UNIT II:**User Research, Personas, and Journey Mapping: User research techniques: Interviews, surveys, contextual inquiry, creating empathy maps and personas, Task flows and user journey mapping, identifying pain points and defining opportunities.

| Sr.No | Questions                                                                                                                                                                                                                                                             | Difficulty Level | Blooms Level | Course Outcome |
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| 1.    | A startup is developing a new mobile application and wants to understand user needs before launch.<br>List any four user research techniques and state what type of data each technique collects.                                                                     | Low              | Remember     | CO2            |
| 2.    | An online shopping app is facing high cart abandonment rates. The team wants to collect user opinions quickly.<br>Explain how surveys can help in this situation. Also identify the type of questions to be included and summarize the insights that can be obtained. | Low              | Understand   | CO2            |
| 3     | A UX team plans to conduct interviews with users for feedback on a prototype.Outline the four main steps involved in conducting effective user interviews and describe each step briefly.                                                                             | Low              | Remember     | CO2            |
| 4     | A bank observes customers using ATMs at different locations to understand their difficulties.Interpret how contextual inquiry works in this case and explain why observing users in a real environment is useful.                                                     | Low              | Understand   | CO2            |
| 5     | A designer is creating an empathy map for users of a food delivery app.Identify the main sections of an empathy map and describe what is included in each section.                                                                                                    | Low              | Remember     | CO2            |
| 6     | A travel booking company creates fictional user profiles based on collected research data.<br>Explain the purpose of personas and illustrate how they help designers make better decisions.                                                                           | Low              | Understand   | CO2            |
| 7     | A team is analyzing how users complete a task in an application step-by-step.Define task flow and list its key components (start point, steps, decision points, end).                                                                                                 | Low              | Remember     | CO2            |
| 8     | A fitness app receives complaints that users find it difficult to track workouts.Explain the concept of pain points and identify two possible pain points from this scenario.                                                                                         | Low              | Understand   | CO2            |
| 9     | A company wants to visualize the complete experience of a user while using their product.Define user journey mapping and list its main elements (stages, actions, emotions, touchpoints).                                                                             | Low              | Remember     | CO2            |

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| 10 | A learning app collects feedback through online forms instead of interviews.Compare surveys and interviews by explaining two advantages of surveys and stating one limitation.                       | Low    | Understand | CO2 |
| 11 | A university library portal is difficult to navigate, and students struggle to find books.Construct a survey by selecting any five questions to collect feedback on search and navigation issues.    | Medium | Apply      | CO2 |
| 12 | Users of a metro ticket booking app report frequent transaction failures.Identify and explain any five pain points faced by users in this process.                                                   | Medium | Analyze    | CO2 |
| 13 | A fitness center is launching a mobile app for its members. Construct a persona including any four elements (e.g., name, goals, frustrations, behavior).                                             | Medium | Apply      | CO2 |
| 14 | A video streaming platform notices users leaving within the first few minutes.Examine and explain any four stages in the user experience where problems may occur.                                   | Medium | Analyze    | CO2 |
| 15 | A railway booking system wants to study how users book tickets during peak hours.Develop a contextual inquiry plan by listing any four steps involved.                                               | Medium | Apply      | CO2 |
| 16 | A mobile game is frequently uninstalled within one day of download.Analyze user behavior and infer any four reasons for this behavior.                                                               | Medium | Analyze    | CO2 |
| 17 | A food court introduces a self-order kiosk for customers.Construct a task flow showing main steps from selecting items to payment (diagram with detailed explanation).                               | Medium | Apply      | CO2 |
| 18 | A hospital collects feedback using forms and face-to-face discussions.Compare these two methods by writing any five differences in the type of insights obtained.                                    | Medium | Analyze    | CO2 |
| 19 | A cab booking service wants to improve the experience for first-time users.construct a user journey by listing any five key steps from app installation to ride completion.(explanation is required) | Medium | Apply      | CO2 |
| 20 | An online news website finds users only read headlines but do not open articles.Categorize and explain any two user behavior patterns causing this issue.                                            | Medium | Analyze    | CO2 |
| 21 | Users of a digital wallet app feel confused while adding money to their account.Identify and explain any four usability problems faced by users in this process.                                     | Medium | Analyze    | CO2 |
| 22 | A ride-sharing app wants to understand daily users like office commuters.Construct a persona including any four elements such as name, goals, frustrations, and behavior.                            | Medium | Apply      | CO2 |

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| 23 | An online job portal finds that users start filling profiles but do not complete them.Examine and explain any five stages where users may face problems.                                                                                            | Medium | Analyze  | CO2 |
| 24 | A supermarket app allows users to scan items and pay without billing counters.Develop a task flow showing the main steps from scanning items to payment confirmation.                                                                               | Medium | Apply    | CO2 |
| 25 | A travel booking website notices users leave the app at the payment stage.Analyze the situation and infer any five reasons for user drop-off.                                                                                                       | Medium | Analyze  | CO2 |
| 26 | A language learning app wants to understand user feedback about new features.Construct an interview plan by writing any four questions or steps.                                                                                                    | Medium | Apply    | CO2 |
| 27 | Users of an online food recipe app complain that they cannot easily follow cooking steps while using the app. Identify and explain any two usability problems that may be causing this issue.                                                       | Medium | Analyze  | CO2 |
| 28 | A smart parking system app allows users to find and reserve parking spots, but users report difficulty in locating available slots.Construct a survey by selecting any five questions to understand issues in searching and booking parking spaces. | Medium | Apply    | CO2 |
| 29 | A public bus tracking app helps users check arrival times of buses.Build a contextual inquiry plan by organizing any four steps to observe how users interact with the app in real situations.                                                      | Medium | Apply    | CO2 |
| 30 | A hospital observes long waiting times for patients even after booking appointments online.Construct any two possible opportunities to improve the patient experience based on this issue.                                                          | Medium | Apply    | CO2 |
| 31 | A government portal needs quick user feedback but has limited time and budget.Select one research method (survey/interview) and justify your choice with any three points.                                                                          | High   | Evaluate | CO2 |
| 32 | A smart home app is designed for elderly users who are not tech-savvy.Design a persona including any four relevant attributes and one key user need.                                                                                                | High   | Create   | CO2 |
| 33 | An e-commerce company conducted interviews but did not improve checkout experience.Assess any five possible issues in their research process.                                                                                                       | High   | Evaluate | CO2 |
| 34 | An online learning platform wants to improve course enrollment.Construct a user journey map showing any four stages with one pain point each.                                                                                                       | High   | Create   | CO2 |
| 35 | A banking app collects large amounts of user data but sees no improvement in user experience.Conclude any Four reasons why the data did not lead to improvements.                                                                                   | High   | Evaluate | CO2 |



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| 36 | A smart city mobile app collects user feedback about public services (water, electricity, roads), but the development team is unsure whether to rely more on surveys or field observations. Evaluate both research methods and justify which method is more effective for identifying real user issues in this scenario (give any three points). | High | Evaluate | CO2 |
| 37 | A telemedicine app allows patients to consult doctors online, but many first-time users feel confused during the consultation process. Construct a user journey map by showing: Any four stages (e.g., login, booking, consultation, payment) One pain point at each stage.                                                                      | High | Create   | CO2 |
| 38 | A smart education platform wants to improve the learning experience for remote students who face difficulty staying engaged. Design and construct a user journey map by organizing: 1. Any four stages of the user experience 2. One pain point and one opportunity at each stage                                                                | High | Create   | CO2 |

| <b>UNIT III: Visual Design Principles and Design Systems</b> UI Design principles: hierarchy, consistency, feedback, accessibility, Typography, color, spacing, iconography, Introduction to Design Systems: Atomic design, Material Design, Wireframing: Low, mid, and high-fidelity wireframes |                                                                                                                                                                                                                                                        |                  |              |                |
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| Sr.No                                                                                                                                                                                                                                                                                            | Questions                                                                                                                                                                                                                                              | Difficulty Level | Blooms Level | Course Outcome |
| 1                                                                                                                                                                                                                                                                                                | A hospital management mobile app shows “Book Appointment” and “Cancel Appointment” buttons with equal size and color, confusing patients. Demonstrate how you would use size, contrast, alignment, and emphasis to establish a clear visual hierarchy. | Low              | Apply        | CO3            |
| 2                                                                                                                                                                                                                                                                                                | A food delivery mobile app uses different button styles on checkout and payment screens. Implement consistency principles by explaining how you will standardize UI elements.                                                                          | Low              | Apply        | CO3            |
| 3                                                                                                                                                                                                                                                                                                | A banking mobile app does not show any response after pressing “Transfer Money”. Construct a feedback system including loading indicators, confirmations, and error messages.                                                                          | Low              | Apply        | CO3            |
| 4                                                                                                                                                                                                                                                                                                | A government healthcare website has low contrast text making it difficult for elderly users to read prescriptions and reports. Utilize accessibility principles to improve readability and visibility.                                                 | Low              | Apply        | CO3            |
| 5                                                                                                                                                                                                                                                                                                | A hospital website displaying lab reports uses very small font and dense text blocks, making it hard for patients to read. Organize typography by defining font size, spacing, and hierarchy.                                                          | Low              | Apply        | CO3            |
| 6                                                                                                                                                                                                                                                                                                | An e-commerce website for medicines uses random colors across product pages, reducing trust and clarity. Develop a consistent color palette including primary, secondary, and accent colors.                                                           | Low              | Apply        | CO3            |
| 7                                                                                                                                                                                                                                                                                                | A news dashboard used by administrators appears cluttered with no spacing between sections like analytics, reports, and alerts. Apply spacing techniques to improve layout clarity.                                                                    | Low              | Apply        | CO3            |
| 8                                                                                                                                                                                                                                                                                                | A mobile health tracking app uses unclear icons for steps, calories, and sleep data, confusing users. Construct meaningful icons ensuring clarity and consistency.                                                                                     | Low              | Apply        | CO3            |
| 9                                                                                                                                                                                                                                                                                                | A hospital software development team repeatedly designs buttons, cards, and forms separately for each module. Apply atomic design to organize reusable UI components.                                                                                  | Low              | Apply        | CO3            |

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| 10 | In a hospital emergency dashboard, critical alerts (e.g., “Patient in ICU needs attention”) appear visually similar to general notifications. Doctors are missing urgent alerts. Examine hierarchy issues and explain how redesigning visual hierarchy will improve attention and response. | Medium | Analyze | CO3 |
| 11 | A travel booking website has different navigation menus on homepage, booking page, and payment page, causing users to get lost. Compare and analyze navigation inconsistencies and suggest a unified navigation structure.                                                                  | Medium | Analyze | CO3 |
| 12 | In a UPI payment app, users are unsure whether the transaction is processing or failed because no clear feedback is shown during delays. Dissect feedback issues and explain how system status, progress indicators, and messages can improve user trust.                                   | Medium | Analyze | CO3 |
| 13 | A medical monitoring app uses only red and green colors to show patient status (critical/normal), making it unusable for color-blind doctors. Differentiate accessibility issues and suggest alternative design solutions.                                                                  | Medium | Analyze | CO3 |
| 14 | A hospital patient portal uses different fonts for headings, reports, and prescriptions, creating confusion in reading medical data. Classify typography issues and explain how a structured typography system improves clarity.                                                            | Medium | Analyze | CO3 |
| 15 | An online pharmacy website uses bright and inconsistent colors across categories like medicines, health products, and offers, reducing readability. Investigate color imbalance and suggest improvements based on harmony and contrast.                                                     | Medium | Analyze | CO3 |
| 16 | A university website displays notices, events, and announcements without proper spacing, making content difficult to scan. Break down spacing issues and explain how spacing improves readability and grouping.                                                                             | Medium | Analyze | CO3 |
| 17 | A fitness tracking mobile app uses different icon styles (flat, 3D, outlined) across screens, confusing users. Inspect inconsistencies and explain how uniform iconography improves usability.                                                                                              | Medium | Analyze | CO3 |
| 18 | A hospital management system has separate UI designs for billing, appointment, and reports modules with no shared components. Analyze and categorize inefficiencies and explain how a design system improves consistency and development speed.                                             | Medium | Analyze | CO3 |

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| 19 | A startup team developing a mobile app directly creates UI screens without wireframes, leading to frequent redesigns and confusion among developers. Examine the issue and explain how low, mid, and high-fidelity wireframes improve design clarity. | Medium | Analyze | CO3 |
| 20 | A news website homepage contains multiple sections (politics, sports, entertainment) but lacks clear grouping and structure, making navigation confusing. Dissect content organization issues and suggest improvements.                               | Medium | Analyze | CO3 |
| 21 | A hospital analytics dashboard displays patient data, reports, and graphs without grouping, causing difficulty in quick decision-making. Analyze layout issues and explain how grouping and alignment improve usability.                              | Medium | Analyze | CO3 |
| 22 | A mobile banking app uses different color schemes for transactions, profile, and dashboard screens, confusing users. Compare inconsistencies and suggest a standardized color system.                                                                 | Medium | Analyze | CO3 |
| 23 | A ride-booking app uses unclear icons for ride history, booking, and payments, causing user errors. Distinguish usability issues and suggest improvements.                                                                                            | Medium | Analyze | CO3 |
| 24 | A ticket booking website does not show confirmation or error messages after user actions like booking or cancellation. Examine feedback issues and suggest improvements.                                                                              | Medium | Analyze | CO3 |
| 25 | A hospital patient mobile app shows different layouts for appointment, reports, and billing sections. Analyze inconsistency issues and explain how design systems improve uniformity.                                                                 | Medium | Analyze | CO3 |
| 26 | A government public service website is not usable by visually impaired users due to poor contrast and missing labels. Investigate accessibility issues and suggest inclusive design improvements.                                                     | Medium | Analyze | CO3 |
| 27 | A financial dashboard used by managers has misaligned charts, tables, and filters, making it difficult to interpret data. Break down alignment issues and suggest grid-based improvements.                                                            | Medium | Analyze | CO3 |
| 28 | A hospital mobile application lacks predefined UI components, leading to inconsistent screens across modules. Analyze the problem and explain how atomic design can structure components effectively.                                                 | Medium | Analyze | CO3 |
| 29 | A mobile application does not follow Material Design guidelines, leading to inconsistent behavior and UI patterns. Examine the                                                                                                                        | Medium | Analyze | CO3 |

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|    | issues and explain how Material Design principles improve usability.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |         |     |
| 30 | A startup is designing a new mobile app and wants to maintain consistency across all screens. Construct an example of Atomic Design structure by organizing components into atoms, molecules, and organisms for a login screen.                                                                                                                                                                                                                                                                                                                | Medium | Analyze | CO3 |
| 31 | A hospital appointment booking mobile app is facing issues where patients are confused while selecting doctors, dates, and time slots due to poor layout and unclear flow. Analyze the existing usability issues and draw a Mid-Fidelity Wireframe showing improved screen layout for doctor selection, date/time selection, and booking confirmation. Clearly label all sections and briefly justify your design choices.                                                                                                                     | Medium | Analyze | CO3 |
| 32 | A banking mobile app prototype developed for user testing includes multiple animations (sliding menus, animated transitions, auto-refreshing graphs) and excessive visual elements on the dashboard screen. During usability testing, users report difficulty in focusing on key tasks like checking balance and transferring money due to distractions and cognitive overload. Analyze the usability issues present in this high-fidelity prototype and suggest suitable simplification strategies to improve task efficiency and user focus. | Medium | Analyze | CO3 |
| 33 | A hospital report interface used by doctors and patients displays medical data such as test results, headings, values, and notes using similar font sizes, weights, and styles. As a result, users struggle to differentiate between section headings, critical values, and normal data while reviewing reports. Examine the typography-related issues in this interface and suggest improvements to establish a clear and effective typography hierarchy.                                                                                     | Medium | Analyze | CO3 |
| 34 | A shopping mobile application displays product details such as name, price, description, and offers using color combinations with low contrast (e.g., light gray text on white background). Users report difficulty in reading product information, especially in different lighting conditions. Investigate the readability and contrast issues in the interface and suggest improvements using appropriate color and accessibility principles.                                                                                               | Medium | Analyze | CO3 |
| 35 | A design team skips wireframing and directly creates high-fidelity designs (Prototype), causing confusion and rework. Analyze the issues and explain the importance of wireframing.                                                                                                                                                                                                                                                                                                                                                            | Medium | Analyze | CO3 |
| 36 | A news portal homepage displays various categories such as                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Medium | Analyze | CO3 |

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|    | breaking news, trending topics, advertisements, and general articles using similar font sizes, colors, and layout styles. Users find it difficult to identify important or urgent news quickly due to lack of emphasis and visual differentiation. Analyze the hierarchy issues in the design and suggest improvements to highlight important content effectively using visual hierarchy principles.                                       |      |          |     |
| 37 | In a hospital appointment app, critical actions like “Emergency Booking” and normal actions like “View History” are given equal visual importance, leading to delays in emergency handling. Judge the effectiveness of the design and justify how hierarchy improvements will enhance usability.                                                                                                                                           | High | Evaluate | CO3 |
| 38 | Evaluate the effectiveness of visual hierarchy in a given UI design by examining the use of its six tools.                                                                                                                                                                                                                                                                                                                                 | High | Evaluate | CO3 |
| 39 | A hospital alert system dashboard uses inconsistent colors for warnings, normal updates, and critical alerts, causing misinterpretation. Critique the design and justify improvements using color principles.                                                                                                                                                                                                                              | High | Evaluate | CO3 |
| 40 | An e-commerce analytics dashboard used by business managers displays sales data, customer insights, and alerts on a single screen without proper grouping, hierarchy, or alignment. Managers struggle to quickly identify important trends and make decisions. Design and develop an improved dashboard layout, clearly explaining how you will apply grouping, alignment, visual hierarchy, and spacing to enhance clarity and usability. | High | Create   | CO3 |
| 41 | Describe the concept of wireframing and discuss various types of wireframes in detail with examples.                                                                                                                                                                                                                                                                                                                                       | High | Evaluate | CO3 |
| 42 | A travel website shows many buttons with the same size, color, and importance, which confuses users about what to click first. Explain and justify how applying visual design principles (such as contrast, hierarchy, alignment, and emphasis) can solve this problem and guide users effectively.                                                                                                                                        | High | Evaluate | CO3 |
| 43 | A university web portal used by students and faculty uses multiple font styles randomly across pages, affecting readability and professionalism. Users find it difficult to distinguish headings, notices, and important information. Construct a typography system defining font hierarchy, font sizes, styles, and usage rules to improve readability and consistency.                                                                   | High | Create   | CO3 |
| 44 | A financial analytics dashboard used by business executives displays key metrics such as revenue, expenses, and performance                                                                                                                                                                                                                                                                                                                | High | Create   | CO3 |

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|    | indicators but lacks alignment, spacing, and structure. Users find it difficult to interpret and compare data quickly. Develop an improved interface layout by applying alignment, spacing, grouping, and visual design principles to enhance clarity and usability..                                                                                                                                                                   |      |        |     |
| 45 | A product team designing a food delivery mobile application needs to present a realistic interface for usability testing and stakeholder approval. The current design lacks detailed structure, interaction flow, and visual clarity, making evaluation difficult. Formulate a high-fidelity wireframe plan specifying layout structure, UI elements, interactions, and visual details required for effective testing and presentation. | High | Create | CO3 |

| <b>UNIT IV: Prototyping, Usability Testing, and UI Development Prototyping using tools (Figma, Penpot), User testing methods: heuristic evaluation, usability testing, Basics of front-end development: HTML, CSS, Design-to-code practices and developer handoff(Anima, Figma Plugins)</b> |                                                                                                                                                                                                        |                  |              |                |
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| Sr. No                                                                                                                                                                                                                                                                                      | Questions                                                                                                                                                                                              | Difficulty Level | Blooms Level | Course Outcome |
| 1                                                                                                                                                                                                                                                                                           | You are designing a login screen in Figma. Your mentor asks you to quickly show how the button will look when clicked. Recall the prototyping feature you would use.                                   | Low              | Remember     | CO4            |
| 2                                                                                                                                                                                                                                                                                           | While testing a mobile app, a user struggles to find the “Submit” button. Explain the kind of usability issue this represents, and describe a method that can be used to identify such problems early. | Low              | Understand   | CO4            |

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| 3  | You are asked to evaluate a website using heuristic evaluation. You notice inconsistent font sizes across pages. Explain the usability principle being violated.                             | Low    | Understand | CO4 |
| 4  | You designed a webpage layout in Figma, and now you need to convert it into a webpage. List the basic front-end technologies you will use.                                                   | Low    | Remember   | CO4 |
| 5  | After writing HTML code for a webpage, you want to improve its visual appearance (colors, spacing). Explain what should be used and justify its purpose.                                     | Low    | Understand | CO4 |
| 6  | You created a responsive design in Figma. While developing it, the layout breaks on smaller screens. Apply the appropriate basic front-end concept to fix this issue.                        | Low    | Apply      | CO4 |
| 7  | You are reviewing a website and notice that similar buttons look different on each page. Explain the usability principle being affected.                                                     | Low    | Understand | CO4 |
| 8  | You want to arrange items in a row with equal spacing using CSS. Name the layout method you can use.                                                                                         | Low    | Remember   | CO4 |
| 9  | You created a multi-screen prototype in Figma, but users are getting confused while navigating between screens. Analyze the prototype and determine modifications to improve user flow.      | Medium | Analyze    | CO4 |
| 10 | You are designing a food ordering app in Figma. Apply interactive prototyping techniques to simulate the complete order placement flow, from menu browsing to order confirmation.            | Medium | Apply      | CO4 |
| 11 | A developer could not accurately replicate the typography and spacing from Figma design files. Analyze how Figma's Inspect panel, design tokens, and plugins could resolve this discrepancy. | Medium | Analyze    | CO4 |
| 12 | Analyze the steps involved in usability testing and examine how each step contributes to identifying UX problems.                                                                            | Medium | Analyze    | CO4 |
| 13 | A usability test found that users took 4 minutes to complete a 3-step checkout. Analyze the design bottlenecks and propose front-end improvements using HTML/CSS best practices.             | Medium | Analyze    | CO4 |
| 14 | You are converting a Figma mockup of a news portal into a responsive HTML/CSS layout. Apply Flexbox or Grid to ensure the design adapts across mobile, tablet, and desktop viewports.        | Medium | Apply      | CO4 |



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| 15 | Your Figma prototype works well on desktop but fails to adapt on mobile screens. Apply appropriate design changes to ensure responsiveness.                                                                             | Medium | Apply   | CO4 |
| 16 | Apply usability testing methodology to design a test plan for a library management system. Include test objectives, participant criteria, task scenarios, success metrics, and a reporting format.                      | Medium | Apply   | CO4 |
| 17 | An e-commerce platform's implemented code deviates from the Figma design because design tokens were not used. Analyze how Figma plugin integrations and design systems can prevent such deviations.                     | Medium | Analyze | CO4 |
| 18 | A mobile banking app has frequent user errors during fund transfers. Apply heuristic evaluation to identify the usability violations and propose an improved interaction design.                                        | Medium | Apply   | CO4 |
| 19 | A UX team created a paper prototype for a smart home app. Analyze what user feedback can be collected at this fidelity level and explain its limitations compared to a high-fidelity Figma prototype.                   | Medium | Analyze | CO4 |
| 20 | You are converting a Figma product listing page into a responsive front-end. Apply CSS Grid to build a 3-column product card layout with a collapsible sidebar filter, describing the HTML structure and key CSS rules. | Medium | Apply   | CO4 |
| 21 | During developer handoff, the developer misinterprets your design. Analyze the documentation or design details that should be improved.                                                                                 | Medium | Analyze | CO4 |
| 22 | While using a Figma plugin like Anima, the generated code is not optimized. Analyze the steps to improve code quality.                                                                                                  | Medium | Analyze | CO4 |
| 23 | Users complain that a form is too long and confusing. Apply appropriate design changes to redesign the prototype and improve usability.                                                                                 | Medium | Apply   | CO4 |
| 24 | During usability testing, users repeatedly make errors while filling a form. Analyze the design and suggest improvements.                                                                                               | Medium | Analyze | CO4 |
| 25 | A developer asks for spacing, color, and typography details. Apply appropriate methods to extract and share this information effectively.                                                                               | Medium | Apply   | CO4 |
| 26 | While using CSS Grid, your layout is not behaving as expected. Analyze the issue and determine steps to fix it.                                                                                                         | Medium | Analyze | CO4 |

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| 27 | Your prototype includes animations that slow down performance. Analyze the changes needed to improve performance.                                                                                                                                                   | Medium | Analyze  | CO4 |
| 28 | In usability testing, users cannot identify clickable elements. Apply appropriate design changes to improve affordance in your design.                                                                                                                              | Medium | Apply    | CO4 |
| 29 | You are handing off a design but assets (icons/images) are missing. Apply appropriate steps to ensure completeness.                                                                                                                                                 | Medium | Apply    | CO4 |
| 30 | A client requests that the UI should be consistent across devices and platforms. Analyze the design and development approach to ensure consistency.                                                                                                                 | Medium | Analyze  | CO4 |
| 31 | During usability testing, users struggle to understand icons without labels. Analyze the design and recommend appropriate changes.                                                                                                                                  | Medium | Analyze  | CO4 |
| 32 | In a heuristic evaluation, it is observed that the system does not provide feedback after user actions. Utilize heuristic evaluation principles and explain how this issue can be identified and addressed to improve user experience.                              | Medium | Apply    | CO4 |
| 33 | Evaluate the effectiveness of Figma's interactive prototyping over traditional paper prototypes for usability testing in an agile environment. Justify your evaluation using at least three criteria: speed, fidelity, and stakeholder communication.               | High   | Evaluate | CO4 |
| 34 | Design a complete prototyping and usability testing plan for a telemedicine app for rural users with limited digital literacy. Include prototype fidelity rationale, five task scenarios, heuristic evaluation criteria, and a post-test iteration strategy.        | High   | Create   | CO4 |
| 35 | Describe the fundamentals of HTML and CSS with examples.                                                                                                                                                                                                            | High   | Evaluate | CO4 |
| 36 | Create a front-end implementation plan for converting a 5-screen Figma prototype of a job portal into a responsive React application. Detail component hierarchy, CSS design token usage, responsive breakpoints, and how the Figma handoff informed each decision. | High   | Create   | CO4 |
| 37 | Evaluate the impact of skipping usability testing in a government healthcare portal. Analyze the consequences for user adoption, error rates, and accessibility compliance, and propose a minimal viable testing strategy under budget constraints.                 | High   | Evaluate | CO4 |

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| 38 | Describe the process of heuristic evaluation using Jakob Nielsen's heuristics. How does it help identify usability issues?                                                                               | High | Evaluate | CO4 |
| 39 | You designed a complex multi-user workflow in Figma, but usability testing reveals users are taking unintended paths. Analyze the information architecture and interactions to optimize task completion. | High | Analyze  | CO4 |
| 40 | During heuristic evaluation, you find multiple violations of usability principles across screens. Evaluate the issues to prioritize and justify which ones to fix first.                                 | High | Evaluate | CO4 |

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